Introduction to Abstract Algebra (In Progress)

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Overview

1. Introduction

This text introduces a mathematical subject called *abstract algebra*.

Before we begin, I would like to thank Alex Tsun for his suggestions and feedback on parts of this material. Also, the inclusion of this overview is heavily influenced by Charles C. Pinter's *A Book of Abstract Algebra*, which is how I first learned this subject.

For this overview, we assume familiarity with real numbers, variables, and equations.

2. What is Abstract Algebra?

Let us do a brief lookback at our previous studies in algebra, to motivate and provide perspective for what is to come.

When we first learned math, we saw how to compute expressions with numbers, such as $2 + 3 = 5$. Then, when we learned algebra, we shifted to representing numbers with letters. This led to the concept of solving an equation for an unknown variable. For example, we would solve $2 + x = 5$ to get $x = 3$. We would then devote our attention to solving all types of equations – linear, quadratic, polynomial, etc. All the quantities involved in these equations were generally real numbers.

But we can in fact do "algebra" with other types of objects. For example, consider the set of all colors, and say we have an operation on colors, denoted by $+$, which

is the result of mixing two colors. So yellow+blue = green, white+black = grey, and so on. We can now do "algebra with colors!" We can write down general properties – for instance, since mixing a color with itself produces just that color, we have, for any color x,

$$x + x = x.$$

Also, it does not matter which order we mix the colors (we define the color mixing operation so that proportions are always the same):

$$a + b = b + a.$$

We can write down (and solve) equations:

$$x + \text{blue} = \text{green} \to x = \text{yellow}.$$

If we compare the general properties satisfied by real numbers and colors, we see that there are some similarities (like $a + b = b + a$) but also some differences (like $x + x = x$.)

When abstract algebra was founded, mathematicians had been studying many other "algebras" like these – in fact, it is estimated that today over 200 algebras have been studied. Every time they wanted to formulate a new algebra, they would model some operations on objects and use those operations to build the main concepts of algebra: expressions, equations, and so forth. Each of these systems was both similar to and different from the others. As more exotic types of objects and operations were studied, the subject of algebra shifted from focusing on the real numbers to embracing the full landscape of all these algebras. Soon after, mathematicians started asking: instead of working in each of these algebras in isolation, could we prove theorems that could apply to multiple algebras at once?

Enter abstract algebra. We can see that many properties of an algebra can be derived from a few "fundamental" properties. For instance, consider the real numbers, which satisfy properties like:

1. $a + b = b + a$ (The *Commutativity of Addition*)

2. $ab = ba$ (The *Commutativity of Multiplication*)
3. $(a + b) + c = a + (b + c)$ (The *Associativity of Addition*)
4. $(ab)c = a(bc)$ (The *Associativity of Multiplication*)
5. The existence of a *neutral element* or *identity* 0 for addition, where $0+r = r + 0 = r$
6. The existence of a similar identity 1 for multiplication, where $1r = r1 = r$
7. The existence, for each element r, of its *negative* or *additive inverse* $-r$, where $r + (-r) = (-r) + r = 0$
8. The existence, for each nonzero element r, of its *reciprocal* or *multiplicative inverse* r^{-1}, where $rr^{-1} = r^{-1}r = 1$ (The "nonzero" is important!)
9. $a(b + c) = ab + ac$ (The *Distributivity of Multiplication Over Addition*)
10. $0r = r0 = 0$

Now, say we were solving the linear equation $ax+b = 0$. Recall that the solution is $x = -b/a = -(ba^{-1})$ if $a \neq 0$. How did we arrive at that solution? Let us lay it out in terms of very small, basic steps:

$ax + b$	$=$	0	Start
$(ax + b) + (-b)$	$=$	$0 + (-b)$	Add $-b$ to Both Sides
$(ax + b) + (-b)$	$=$	$-b$	$0 + r = r + 0 = r$ for all r
$ax + (b + (-b))$	$=$	$-b$	Associativity of Addition
$ax + 0$	$=$	$-b$	$r + (-r) = (-r) + r = 0$ for all r
ax	$=$	$-b$	$0 + r = r + 0 = r$ for all r
$a^{-1}(ax)$	$=$	$a^{-1}(-b)$	Multiply Both Sides by a^{-1}, $a \neq 0$
$(a^{-1}a)x$	$=$	$a^{-1}(-b)$	Associativity of Multiplication
$1x$	$=$	$a^{-1}(-b)$	$rr^{-1} = r^{-1}r = 1$ for all r
x	$=$	$a^{-1}(-b)$	$1r = r1 = r$ for all r
x	$=$	$(-b)a^{-1}$	Commutativity of Multiplication
$x + ba^{-1}$	$=$	$(-b)a^{-1} + ba^{-1}$	Add ba^{-1} to Both Sides
$x + ba^{-1}$	$=$	$((-b) + b)a^{-1}$	Distributivity

$x + ba^{-1}$	$=$	$0a^{-1}$	$r + (-r) = (-r) + r = 0$ for all r
$x + ba^{-1}$	$=$	0	$0r = r0 = 0$ for all r
$(x + ba^{-1}) + (-ba^{-1})$	$=$	$0 + (-ba^{-1})$	Add $-ba^{-1}$ to Both Sides
$x + (ba^{-1} + (-ba^{-1}))$	$=$	$0 + (-ba^{-1})$	Associativity of Addition
$x + 0$	$=$	$0 + (-ba^{-1})$	$r + (-r) = (-r) + r = 0$ for all r
x	$=$	$-(ba^{-1})$	$0 + r = r + 0 = r$ for all r

Now, the fundamental idea that enables abstract algebra is this: we do not actually need a, x, or b to be real numbers for this process to work. All we need are those properties to be satisfied!

In other words, if we assume that we generally had some set of mathematical objects – they might be real numbers, or they might not be – together with some operations on those objects, that satisfied the properties above, then we can solve linear equations $ax + b = 0$ for those objects, the same way we solve linear equations for real numbers. In fact, anything we do in this general setting can immediately carry over to any algebra that shares these properties with the real numbers. We *abstract* the fact that the objects and operations satisfy the properties away from what the objects or operations actually are, hence the term, *abstract algebra*.

To give an example of this process, consider the generalization of the algebra of real numbers to this concept:

Definition 2.1. A *field* is a set F together with operations $+, \times$ on the elements of F. These operations must be *binary*: they both must take in two elements and output one element. They must also satisfy the following properties:

1. *Commutativity*: For all a, b in F, $a + b = b + a$ and $a \times b = b \times a$

2. *Associativity*: For all a, b, c in F, $(a + b) + c = a + (b + c)$ and $(a \times b) \times c = a \times (b \times c)$

3. *Additive and multiplicative identity*: There exist elements of F called 0 and 1, such that, for all a in F, $0 + a = a + 0 = a$ and $1 \times a = a \times 1 = a$
4. *Additive inverse*: For all a in F, there exists an element $-a$ such that $a + (-a) = (-a) + a = 0$
5. *Multiplicative inverse*: For all *nonzero* a in F, there exists an element a^{-1} such that $a \times a^{-1} = a^{-1} \times a = 1$
6. *Distributivity*: For all a, b, c in F, $(a + b) \times c = (a \times c) + (b \times c)$ and $a \times (b + c) = (a \times b) + (a \times c)$

From these defining properties of a field, we can actually prove $0x = x0 = 0$ as a consequence. Thus, linear equation solutions for general fields are the same as linear equation solutions for real numbers. We can go on to ask how much real number algebra can carry over – and indeed we can discuss polynomials over fields and theorems for fields that heavily generalize theorems about real numbers.

We can then identify many sets of objects which these theorems can apply to. For example, consider the set of just 0 and 1 with the following "addition" and "multiplication" operations:

$$\begin{aligned}
0 + 0 &= 0, \\
0 + 1 &= 1, \\
1 + 0 &= 1, \\
1 + 1 &= 0, \\
0 \times 0 &= 0, \\
0 \times 1 &= 0, \\
1 \times 0 &= 0, \\
1 \times 1 &= 1.
\end{aligned}$$

This is actually a field! (It is called $\mathbb{Z}_2$.) In fact, this is a special case of a more general kind of construction. Consider say the set of numbers from 0 to 9,

with the operations of usual addition and multiplication but "wrapped around" to stay in 0 through 9 (so for example $9 + 9 = 18 \to 18 - 10 = 8$.) In other words, the operations are the remainders when divided by 10 of usual addition and multiplication. We can do this for any number $n > 1$: we define $\mathbb{Z}_n$ to be the set of numbers from 0 to $n - 1$ (including both) with the operations as remainders when divided by n of usual addition and multiplication. Then, it turns out that if n is prime, $\mathbb{Z}_n$ is a field! We will prove this in general later on. In fact, these "finite fields" turn out to have lots of interesting applications, including being foundational to modern computer security.

We can alternatively study a different set of defining properties. For example, another concept in abstract algebra is a *group*:

Definition 2.2. A *group* is a set G together with a binary operation $\times$ satisfying:

1. *Associativity*: For all a, b, c in G, $(a \times b) \times c = a \times (b \times c)$
2. *Identity*: There exists an element of G, which we usually call e, such that, for all g in G, $g \times e = e \times g = g$
3. *Inverse*: For all g in G, there exists an element g^{-1} such that $g \times g^{-1} = g^{-1} \times g = e$

This is just a subset of the defining properties of fields. This automatically means that all fields are groups, but so are many other algebras that aren't fields – and all of group theory applies to all of these cases. As an example, it turns out that the set of transformations you can apply to a Rubik's cube is a group – in fact, group theory is an important tool for discovering algorithms to solve Rubik's cubes quickly[1].

These abstracted concepts – fields and groups – are called *algebraic structures*. An algebraic structure is essentially any defining list of properties that a set of operations are required to satisfy.

The strength of abstract algebra is many-fold. First, theorems on algebraic

[1] The application of group theory to Rubik's cubes is not discussed in this text, but if you're interested you should look into it afterwards!

structures extend results on familiar algebras like the real numbers to other sets of objects. But in the other direction, studying abstract algebra has actually resulted in new theorems even for familiar algebras like the real numbers! One way this can happen is that abstract algebra allows you to focus on the consequences of a particular set of properties, enabling you to discover results that you may have missed before. We will see more examples of this phenomenon later on.

3. The Rest of the Text

We now discuss some important points to keep in mind for the rest of the text.

In general, we assume a prerequisite of precalculus at the level of Art of Problem Solving's *Precalculus*. In particular, we assume familiarity with algebra (polynomials, functions, etc.), linear algebra (vectors, matrices, etc.), and complex numbers. We always assume comfortability with proofs and mathematical rigor. It is helpful to know some elementary discrete math (e.g., number theory and combinatorics); we also include an appendix where we cover some relevant discrete math background. In some places, we refer to axiomatic set theory (in particular the Axiom of Choice), but we will explicitly mention that beforehand, and those particular proofs can be skipped without loss of continuity if you are willing to accept the corresponding results on faith.

We assume familiarity with standard and common mathematical notation. For example:

1. $\mathbb{R}$ denotes the set of real numbers, $\mathbb{Z}$ the set of integers, and $\mathbb{Q}$ the set of rational numbers.

2. For either of these three sets, attaching the subscript $+$ produces the subset of positive members, so for example $\mathbb{Z}_+$ is the set of positive integers.

3. Similarly, for either of these three sets, attaching the subscript 0 produces the subset of nonnegative members, so for example $\mathbb{R}_0$ is the set of nonnegative real numbers.

4. For any two sets A and B, $A \times B$ denotes the set of ordered pairs where the first "coordinate" is from A and the second is from B. For example, if $A = \{1, 2\}$ and $B = \{2, 3\}$, then $A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}$.

5. In particular, for any set S, $S^2 = S \times S$ denotes the set of ordered pairs of elements of S, where order matters, so for example for $S = \mathbb{R}$, $(2, 3)$ is different from $(3, 2)$. Thus, $S^2 = \{(a, b) : a, b \in S\}$. As another example, if $S = \{1, 2\}$, then $S^2 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.

6. For any three sets A, B, and C, we can talk about $(A \times B) \times C$ or $A \times (B \times C)$, but there is a "natural" way to associate every element of $(A \times B) \times C$ uniquely with an element of $A \times (B \times C)$, so there is no ambiguity if we write $A \times B \times C$.

7. In particular, for any set S, $S^3 = S \times S \times S$ is the set of ordered triples of elements of S, $S^3 = \{(a, b, c) : a, b, c \in S\}$.

8. In general, for any n sets $A_1, ..., A_n$, we can write $A_1 \times ... \times A_n$ without ambiguity.

9. In particular, for any set S, S^n, or S $\times$'d with itself n times, is the set of ordered n-tuples of elements of S, $S^n = \{(a_1, ..., a_n) : a_i \in S\}$.

10. For $x \in A_1 \times ... \times A_n$, x_i denotes the ith "coordinate" of x, which would be an element of A_i. So for example, if $x = (4, 5, 6) \in \mathbb{R}^3$, then $x_1 = 4$, $x_2 = 5$, and $x_3 = 6$.

Many (if not most) abstract algebra curricula start with something other than fields; however, here we proceed in order of fields, vector spaces, groups, and rings. We think that the order should not affect the learning experience too strongly, and fields may be a more "familiar" algebraic structure to start with because of students' experience with rational, real, and complex numbers.

This text is divided naturally into three parts:

1. The first covers two major algebraic structures, fields and vector spaces, and it can serve as a self-contained introduction to their study. This provides ample background to start studying some applications of abstract

algebra, such as vector space analysis (which has its own major applications such as quantum mechanics.)

2. The second covers two additional major algebraic structures, groups and rings, completing the introduction of the four main algebraic structures that commonly comprise an intro to abstract algebra course.

3. The third covers some major topics that provide further results on and applications of these four algebraic structures. Together with the first two parts, these topics complete a typical intro course.

The "fourth part" is an appendix, which as mentioned previously covers some relevant discrete math background.

We do not include any problems or exercises. We think that there are already many great problem compilations out there, and if you want practice, you should look at those places. We recommend that, in addition to standard "plug-and-chug" exercises that may involve just applying formulas in obvious ways, you also look at "brainteaser" problems that involve more creative combinations of applications of theorems. The Putnam Mathematical Competition is a famous college-level extracurricular contest that features these kinds of "brainteaser" problems for subjects that are typically taught in undergraduate math, including abstract algebra.

4. Final Preliminaries

I would appreciate any comments, feedback, or corrections. You can contact me at `https://riemannzeta5.com/contact`.

Now, without further ado, let's begin!

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Part I.

Fields and Vector Spaces

1. Fields

To be completed.